

INDR 100

INTRODUCTION TO INDUSTRIAL ENGINEERING

Lecture #9 Layout Types

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Agenda of the Day

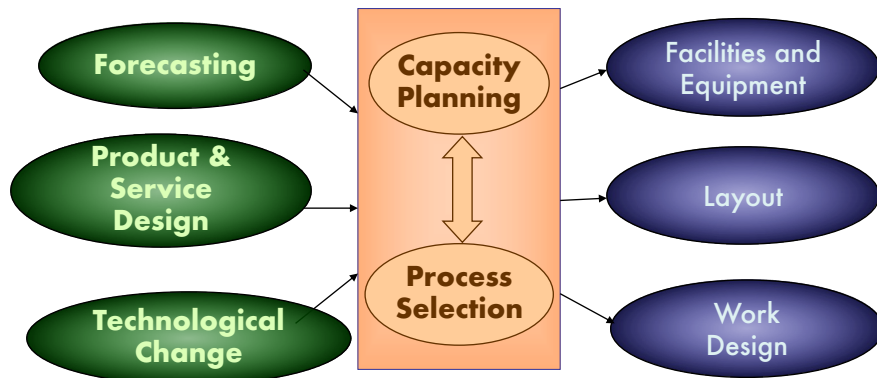
- Processes and layout
- Layout types
 - Product layout
 - Process layout
 - Fixed-position layout
- Class exercise

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Process Selection and System Design



Capacity is significantly impacted by process selection and facility layout.

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Process Selection → Layout

- ❑ Production volume and product variety determine the type of the physical arrangement of the production system, called the **layout** of the system.
- ❑ **Production volume** refers to the number of units of a given part or product produced annually by the plant
- ❑ **Product variety** refers to the different product designs or types that are produced in a plant.

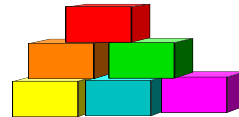
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Facilities Layout

- ❑ The arrangement of departments, work centers, and equipment, with particular emphasis on movement of work (customers or materials) through the system.
- ❑ Objectives include:
 - quickest flow of material
 - lowest cost
 - least amount of handling in processing the product from the receipt of raw material to the delivery of the final product.



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Basic Layout Types

- ❑ **Product Layouts** most helpful to repetitive processing
- ❑ **Process Layouts** used for irregular processing
- ❑ **Fixed-position layouts** used when projects require layouts
- ❑ **Hybrid layouts** combinations of these above types
 - Cellular manufacturing
 - Group technology
 - Flexible Manufacturing Systems

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Product Layouts

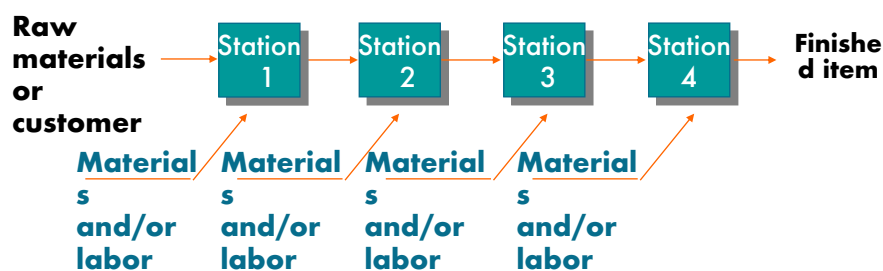
- ❑ **Product layout:** Layout that uses standardized processing operations to achieve smooth, fast, high-volume flow
- ❑ Made possible by highly standardized goods or services that allow highly standardized, repetitive processing
- ❑ Work is divided into a series of standardized tasks, permitting specialization of equipment and division of labor
- ❑ The large volumes handled by these systems usually make it economical to invest substantial sums of money in equipment and in job design.

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Production/Assembly Line

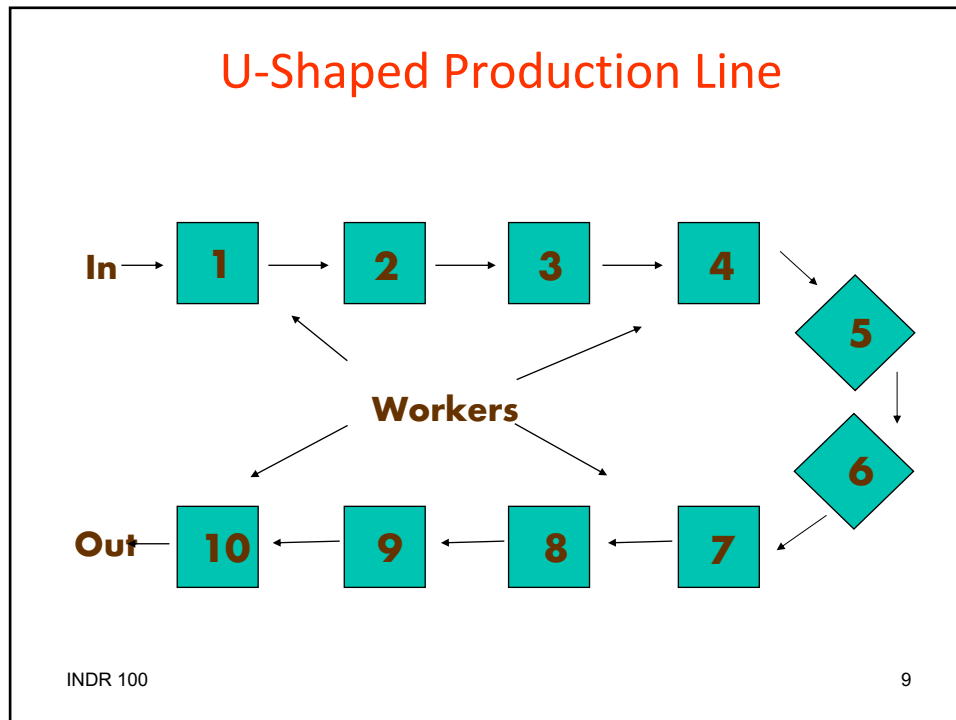


Used for Flow (Repetitive or Continuous) Processing
Example: automobile assembly lines, cafeteria serving line

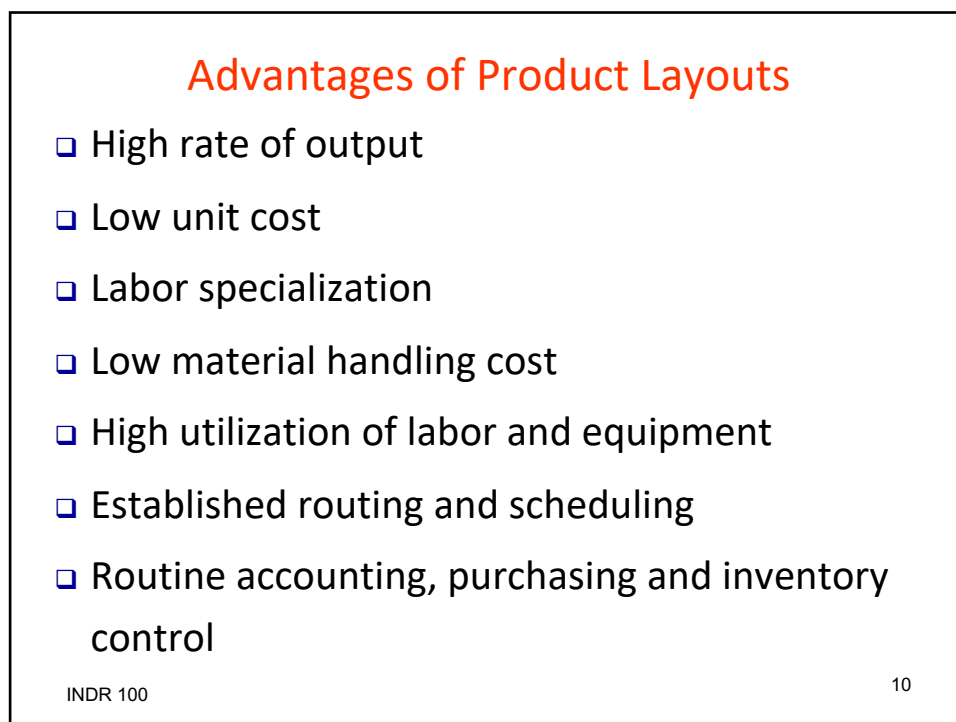
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Disadvantages of Product Layouts

- ❑ Creates boring, repetitive jobs
- ❑ Poorly skilled workers may not maintain equipment or quality of output
- ❑ Fairly inflexible to changes in volume
- ❑ Highly inclined to shutdowns
- ❑ Needs preventive maintenance
- ❑ Individual incentive plans are impractical

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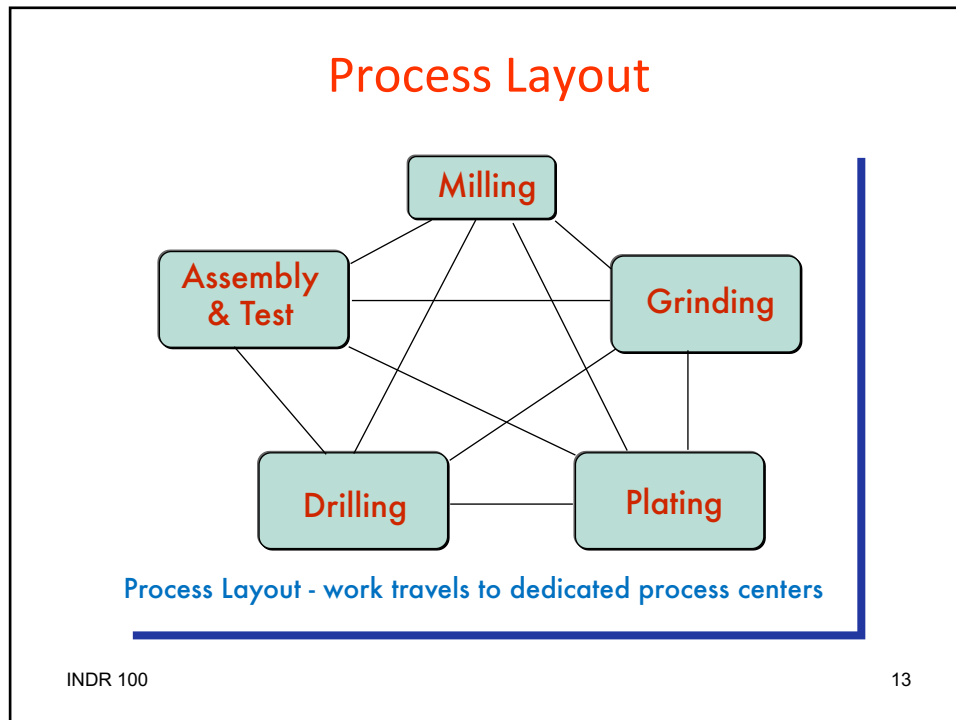
Process Layouts

- ❑ **Process layouts:** Layouts that can handle various processing requirements
- ❑ The layouts feature departments or other functional groupings in which similar kinds of activities are performed
- ❑ Examples: Machine shops usually have separate departments for milling, grinding, drilling, and so on
- ❑ Different products may present quite different processing requirements and sequences of operations

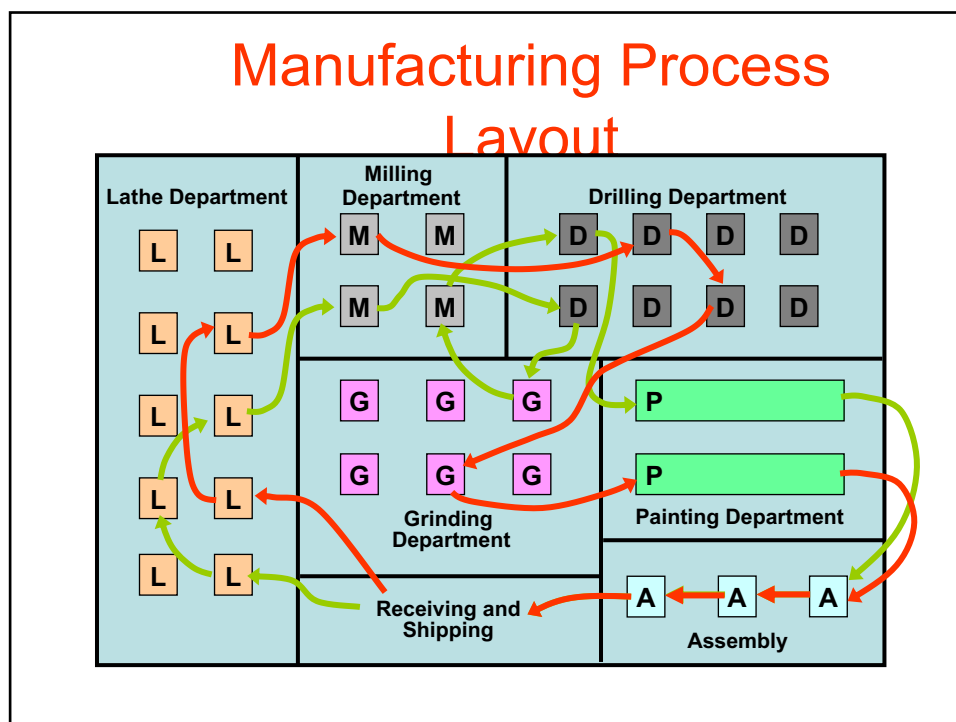
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Process-Oriented Layout

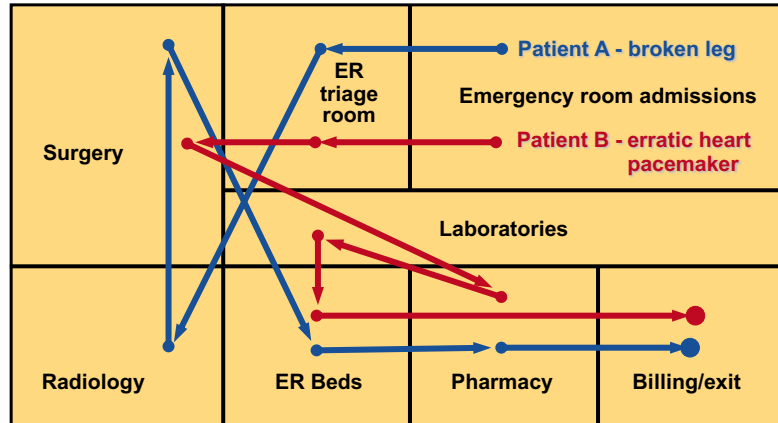


Figure 9.3

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Gölcük Fabrikası – Yeni Nesil Ford Transit'in ana üretim merkezi

The image shows the Gölcük Factory, the main production center for the new Ford Transit. The factory is divided into four main production areas, each color-coded:

- MONTAJ (Green):** Assembly area.
- BOYAHANE (Blue):** Painting area.
- KAYNAK (Red):** Welding area.
- TAKIM KALIP & PRES (Yellow):** Tooling, casting, and pressing area.

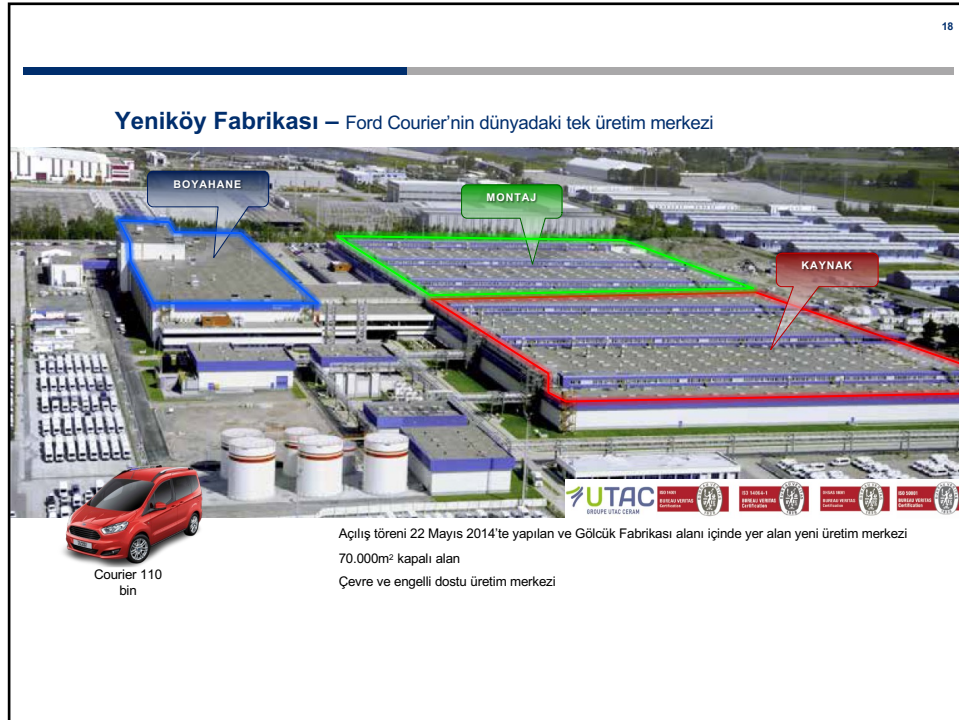
Below the aerial view, two Ford Transit vans are shown: the Transit 140 bin and the Custom 150 bin.

UTAC (GRUPE UTAC CERAR) is the manufacturer.

Açılış: 2001
340.000m² kapalı alan olmak üzere 1.600.000m² toplam alan

Below the text, there are several certification logos, including ISO 9001, ISO 14001, and others.

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Advantages of Process Layouts

- ❑ Can handle a variety of processing requirements
- ❑ Not particularly at risk to equipment failures
- ❑ Equipment used is less costly
- ❑ Possible to use individual incentive plans

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Disadvantages of Process Layouts

- ❑ In-process inventory costs can be high
- ❑ Challenging routing and scheduling
- ❑ Equipment utilization rates are low
- ❑ Material handling slow and inefficient
- ❑ Complexities often reduce span of supervision
- ❑ Special attention for each product or customer
- ❑ Accounting, inventory control and purchasing are more involved

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Fixed-Position Layouts

- ❑ **Fixed-Position Layout:** Layout in which the product or project remains stationary, and workers, materials, and equipment are moved as needed

Examples:

- Large construction projects (buildings, power plants, dams)
- Shipbuilding, production of large aircraft
- Rockets used to launch space missions

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Combination Layouts

- ❑ The three basic layout types may be altered to satisfy the needs of a particular situation
- ❑ **Examples:**
 - Supermarket layouts: primarily process layout, have fixed-path material-handling devices as well (roller-type conveyors and belt-type conveyors)
 - Hospitals: process layout, fixed-position layout as well (patient care)
 - Off-line reworking (customized processing) of faulty parts in a product layout

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Combination Layouts -Example

A small company produces three types of backpacks:

- ❑ School backpack
- ❑ Hiking backpack
- ❑ Laptop backpack

Each requires:

- ❑ Cutting fabric
- ❑ Sewing front pocket
- ❑ Attaching zippers
- ❑ Installing shoulder straps
- ❑ Final stitching and inspection

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Combination Layouts -Example

Traditional Layout (Job Shop)

- ❑ Machines and workers are arranged by function:
 - All fabric cutting in one area
 - All zipper attachment in another
 - All sewing machines together
- ❑ Bags zig-zag through the entire facility, creating:
 - long travel distances
 - large WIP
 - blocking and waiting
 - slow throughput

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Combination Layouts -Example

Cellular Layout Solution

The company reorganizes into **three product cells**.

- ❑ **Cell A: School Bags** → Workers build a school backpack from start to finish inside this cell.
 - Machines:
 - 1 cutting table
 - 2 sewing stations
 - 1 zipper station
 - 1 strap assembly station
- ❑ **Cell B: Hiking Bags** → Similar equipment but arranged for the extra pocket and thicker fabric.
- ❑ **Cell C: Laptop Bags** → Includes an added foam insertion station and different zipper equipment.

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From Capacity to Layout

Facility Example	Capacity Goal	Best Layout Type	Why?
Small (Boutique/Custom)	High Flexibility	Process Layout	Groups similar skills (e.g., a custom woodshop); handles unique, low-volume orders.
Medium (Batch/Hybrid)	Balance	Cellular Layout	Groups different machines into "cells" to act like a mini-assembly line for specific product families.
Large (Mass Production)	High Volume	Product Layout	The classic "Assembly Line"; organized by the sequence of steps to maximize speed.